AVAIL: availability-aware multimodal integration

Every subject contributes; absence is represented, not discarded or fabricated.

DATA

Clinical / cognitive
age · MMSE · UPDRS

16S / shotgun
taxa · pathways

Metabolomics
serum / stool

Imaging
FC · radiomics

Missingness is data
flags carry the pattern

Stage 1 · specialists
one learner / modality

trained only on subjects with that modality

p_im = f_m(x_im)

Availability encoding
z_i = [p_i1, a_i1, ..., p_iM, a_iM]

a_im in {0, 1}

Stage 2 · fusion
meta-learner g(theta)

11-model zoo

q_i = g(z_i)

Decision + report
subject risk score

coverage · calibration
figures · R export

Validation envelope

the same pipeline is exercised at every path — no post-hoc selection

Leak-free evaluation
5-fold / LOO
fold-internal preprocessing

Leakage control
full-pipeline label
permutation null

Missingness stress
MCAR / MAR / MNAR
block dropout

External cohorts
MetaCardis · HMP2
ds004392 · coverage audit